INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Homework 2 Solutions

1. (*6 pts*) A dental clinic is hiring. It needs a receptionist and an office manager, chosen from 6 applicants for the front office, and 2 hygienists with no distinct roles, chosen from 5 applicants for the hygienist positions. No applicant can fill more than one position. In how many ways can the positions be filled?

Solution: There are ${}_6P_2$ ways to fill the receptionist and office manager positions, and for each way there are ${}_5C_2$ ways to fill the hygienist positions. Therefore, there are

$$ {}_6P_2 \cdot {}_5C_2 = 30 \cdot 10 = 300 $$

ways to fill the positions.

2. (*6 pts*) A hospital visitor parking pass number is either a letter (A–Z) followed by two digits (0–9), such as B07, or three digits, such as 315. Letters and digits may be repeated. How many different parking pass numbers are possible?

Solution: By the Fundamental Principle of Counting, for the first type there are

$$ 26 \cdot 10 \cdot 10 = 2{,}600 $$

possible passes. For the second type, there are

$$ 10 \cdot 10 \cdot 10 = 10^3 = 1{,}000 $$

possible passes. The two types are disjoint, so by the Addition Principle there are a total of

$$ 2{,}600 + 1{,}000 = 3{,}600 $$

possible parking passes.

3. (*6 pts*) A die is rolled 3 times. How many of the possible sequences of results include at least one 6?

Solution: The universal set U here is the set of sequences of 3 results, each a whole number from 1 to 6, which has cardinality $|U| = 6 \cdot 6 \cdot 6 = 216$. Let A be the subset of U consisting of sequences with at least one 6. Then A^{c} is the subset of U consisting of sequences with no 6. Each roll can then only land on 5 numbers, so the cardinality of A^{c} is $|A^{\mathrm{c}}| = 5 \cdot 5 \cdot 5 = 125$. Therefore, $|A| = |U| - |A^{\mathrm{c}}| = 216 - 125 = 91$.

4. (*6 pts*) A clinic has 40 staff members. Of these, 22 speak Spanish, 15 speak Polish, and 6 speak both.

(a) How many of the staff members speak at least one of the two languages?

Solution: Let S be the set of staff members who speak Spanish and P the set of staff members who speak Polish. We are given $|S| = 22$, $|P| = 15$, and $|S \cap P| = 6$. Therefore,

$$|S \cup P| = |S| + |P| - |S \cap P| = 22 + 15 - 6 = 31.$$

(b) How many speak neither language?

Solution: If U is the set of all staff members at the clinic, then $|U| = 40$, and

$$\left|(S \cup P)^{\mathrm{c}}\right| = |U| - |S \cup P| = 40 - 31 = 9.$$

5. (*6 pts*) A committee of 4 will be chosen from 5 doctors and 6 nurses, and the committee members have no distinct roles. How many different committees include at least one doctor?

Solution: The total number of committees without the requirement that there be at least one doctor is ${}_{11}C_4 = 330$. The number of committees with no doctors is ${}_6C_4 = 15$. Therefore, the number of committees with at least one doctor is $330 - 15 = 315$.

6. (*14 pts*) A respiratory care team of 5 will be chosen from 6 nurses and 4 respiratory therapists, and the team members have no distinct roles.

(a) (*2 pts*) How many different teams are possible in all?

Solution: There are a total of $6 + 4 = 10$ people in the pool and we must choose 5 of them. Order doesn't matter, so there are ${}_{10}C_5 = 252$ possible teams.

(b) (*3 pts*) How many teams include exactly 2 respiratory therapists?

Solution: There are ${}_4C_2 = 6$ ways to choose the respiratory therapists and then ${}_6C_3 = 20$ ways to choose the remaining team members. Therefore there are $6 \cdot 20 = 120$ teams with exactly 2 respiratory therapists.

(c) (*4 pts*) Split the teams into cases to find the number of teams that include at least 2 respiratory therapists.

Solution: There are 120 teams with exactly 2 respiratory therapists. There are ${}_4C_3 \cdot {}_6C_2 = 4 \cdot 15 = 60$ teams with exactly 3 respiratory therapists, and ${}_4C_4 \cdot {}_6C_1 = 1 \cdot 6 = 6$ teams with all four respiratory therapists. The cases are disjoint, so by the Addition Principle, the number of teams with at least 2 respiratory therapists is $120 + 60 + 6 = 186$.

(d) (*3 pts*) Find the number of teams that include at least 2 respiratory therapists again, this time using complements.

Solution: There are ${}_4C_0 \cdot {}_6C_5 = 1 \cdot 6 = 6$ teams with no respiratory therapists and ${}_4C_1 \cdot {}_6C_4 = 4 \cdot 15 = 60$ teams with exactly one respiratory therapist. Therefore there are $6 + 60 = 66$ teams with 0 or 1 respiratory therapists. Using complements, the number of teams with two or more respiratory therapists is $252 - 66 = 186$.

(e) (*2 pts*) Explain why the methods in parts (c) and (d) must give the same answer.

Solution: Both methods count the teams with at least 2 respiratory therapists, so they must give the same answer.

7. (*14 pts*) A nurse will work 7 shifts next week, and each shift is either a day shift (D) or a night shift (N). A schedule for the week is a sequence such as DDNDNDD.

(a) (*2 pts*) How many different schedules are possible?

Solution: Each of the 7 shifts is either a day shift or a night shift, so by the Fundamental Principle of Counting there are $2^7 = 128$ possible schedules.

(b) (*2 pts*) How many schedules have exactly 2 night shifts?

Solution: A schedule with exactly 2 night shifts is determined by choosing which 2 of the 7 shifts are night shifts, so there are ${}_7C_2 = 21$ such schedules.

(c) (*2 pts*) How many schedules have at least one night shift?

Solution: The number of schedules with no night shifts is 1 – all day shifts, corresponding to the sequence DDDDDDD. Therefore the number of schedules with at least one night shift is $128 - 1 = 127$.

(d) (*3 pts*) How many schedules have at least 2 night shifts?

Solution: There is 1 schedule with no night shifts and ${}_7C_1 = 7$ schedules with exactly one night shift. Therefore, the number of schedules with at least two night shifts is $128 - 1 - 7 = 120$.

(e) (*3 pts*) How many schedules have at most 5 night shifts?

Solution: The number of schedules with exactly 6 night shifts is ${}_7C_6 = 7$ and the number of schedules with exactly 7 night shifts is ${}_7C_7 = 1$. Therefore the number of schedules with at most 5 night shifts is $128 - 7 - 1 = 120$.

(f) (*2 pts*) Your answers to parts (d) and (e) should be equal. Explain why.

Solution: The number of schedules with no night shifts is ${}_7C_0 = 1$, which is the same as ${}_7C_7$, the number of schedules with all 7 night shifts. Similarly, the number of schedules with exactly one night shift is ${}_7C_1 = 7$, which is the same as ${}_7C_6$, the number of schedules with exactly 6 night shifts. So, excluding the schedules with 0 or 1 night shifts is the same as excluding the schedules with 6 or 7 night shifts.

8. (*14 pts*) Let A be the set of whole numbers from 1 to 200 that are divisible by 4, and let B be the set of whole numbers from 1 to 200 that are divisible by 6.

(a) (*2 pts*) Find $|A|$.

Solution: $|A| = \left\lfloor \frac{200}{4} \right\rfloor = 50$.

(b) (*2 pts*) Find $|B|$.

Solution: $|B| = \left\lfloor \frac{200}{6} \right\rfloor = 33$.

(c) (*4 pts*) A student says that $A \cap B$ is the set of multiples of 24, since $4 \cdot 6 = 24$. Explain why the student is wrong, and find $|A \cap B|$.

Solution: The least common multiple of 4 and 6 is 12, not 24. All whole numbers that are divisible by 4 and 6 are also divisible by 12. So, $|A \cap B| = \left\lfloor \frac{200}{12} \right\rfloor = 16$.

(d) (*3 pts*) How many whole numbers from 1 to 200 are divisible by 4 or by 6?

Solution: $|A \cup B| = |A| + |B| - |A \cap B| = 50 + 33 - 16 = 67$.

(e) (*3 pts*) How many whole numbers from 1 to 200 are divisible by neither 4 nor 6?

Solution: $200 - 67 = 133$.

9. (*14 pts*) A hospital gives each patient an ID made of two letters (A–Z) followed by two digits (0–9), such as MM47. Letters and digits may be repeated.

(a) (*2 pts*) How many different IDs are possible?

Solution: $26 \cdot 26 \cdot 10 \cdot 10 = 67{,}600$.

(b) (*3 pts*) How many IDs have two different letters and two different digits?

Solution: There are 26 choices for the first letter, 25 for the second letter, 10 for the first digit, and 9 for the second digit, so by the Fundamental Principle of Counting there are $26 \cdot 25 \cdot 10 \cdot 9 = 58{,}500$ such IDs.

(c) (*3 pts*) Use complements and your answers to parts (a) and (b) to find the number of IDs in which the two letters are the same or the two digits are the same.

Solution: $67{,}600 - 58{,}500 = 9{,}100$.

(d) (*4 pts*) Let L be the set of IDs in which the two letters are the same, and let D be the set of IDs in which the two digits are the same. Find $|L|$, $|D|$, and $|L \cap D|$, and use the Inclusion-Exclusion Principle to find $|L \cup D|$.

Solution: Once the first letter is chosen, the second letter is forced, so $|L| = 26 \cdot 10 \cdot 10 = 2{,}600$. Similarly, once the first digit is chosen, the second digit is forced, so $|D| = 26 \cdot 26 \cdot 10 = 6{,}760$. In $L \cap D$, both the second letter and the second digit are forced, so $|L \cap D| = 26 \cdot 10 = 260$. By the Inclusion-Exclusion Principle,

$$|L \cup D| = |L| + |D| - |L \cap D| = 2{,}600 + 6{,}760 - 260 = 9{,}100.$$

(e) (*2 pts*) Your answers to parts (c) and (d) should be equal. Explain why.

Solution: Both methods count the IDs in which the two letters are the same or the two digits are the same, so they must give the same answer.

10. (*14 pts*) A hospital unit has 10 nurses, including Rosa and Sam, and 6 nursing aides. Each shift needs a charge nurse, who must be one of the nurses, and 4 other team members with no distinct roles, chosen from the remaining nurses and aides. The 4 other team members must include at least one aide. In addition, Rosa and Sam refuse to work the same shift. In how many ways can a shift be staffed?

(a) (*2 pts*) Ignore the requirement about aides and Rosa and Sam's refusal to work together. In how many ways can the shift be staffed?

Solution: There are ${}_{10}C_1 = 10$ ways to choose the charge nurse and then ${}_{15}C_4 = 1{,}365$ ways to choose the 4 team members out of the remaining 15 people, for a total of $10 \cdot 1{,}365 = 13{,}650$.

(b) (*2 pts*) Still ignoring Rosa and Sam's refusal to work together, in how many ways can the shift be staffed with no aides among the 4 other team members?

Solution: There are ${}_{10}C_1 = 10$ ways to choose the charge nurse and then ${}_{9}C_4 = 126$ ways to choose the 4 team members out of the remaining 9 nurses, for a total of $10 \cdot 126 = 1{,}260$.

(c) (*2 pts*) Use your answers to parts (a) and (b) to find the number of ways to staff the shift with at least one aide, still ignoring Rosa and Sam's refusal to work together.

Solution: Subtract the number of teams with no aide from the number found in part (a): $13{,}650 - 1{,}260 = 12{,}390$.

(d) (*3 pts*) How many ways are there to staff the shift with at least one aide, if Rosa is the charge nurse and Sam is one of the 4 other team members? How many are there if Sam is the charge nurse and Rosa is one of the 4 other team members?

Solution: There are ${}_{14}C_3 = 364$ ways to choose the other 3 team members from the remaining 14 people; however, this counts teams with no aides, of which there are ${}_8C_3 = 56$. Therefore there are $364 - 56 = 308$ such teams. The same is true when Sam is the charge nurse and Rosa is one of the 4 team members.

(e) (*3 pts*) How many ways are there to staff the shift with at least one aide, if Rosa and Sam are both among the 4 other team members?

Solution: There are ${}_8C_1 = 8$ ways to choose the charge nurse from the remaining 8 nurses. There are ${}_{13}C_2 = 78$ ways to choose the remaining 2 team members; however, this counts teams with no aide, of which there are ${}_7C_2 = 21$. Therefore, there are $8 \cdot (78 - 21) = 8 \cdot 57 = 456$ such teams.

(f) (*2 pts*) Use your answers to parts (c), (d), and (e) to answer the original question. Which counting principles did you use?

Solution: In parts (d) and (e) we counted the teams with both Rosa and Sam and at least one aide. If both Rosa and Sam are on the shift, then either one of them is the charge nurse or both are among the 4 other team members, so these three cases are disjoint and cover every such team. By the Addition Principle, we can subtract all three numbers from the number found in part (c) to obtain the final count: $12{,}390 - 308 - 308 - 456 = 11{,}318$. We used the Fundamental Principle of Counting, combinations, complements, and the Addition Principle.